KATHERINE A. BARBIERI

Totowa, NJ | (845)527-2304 | katherineabarbieri@gmail.com

PROFESSIONAL SUMMARY

Dynamic and motivated biomedical engineering professional with over six years of Quality/R&D experience; specializing in technical report writing, data analysis, test method development, design verification/validation for sustaining projects and new product development/product scale-up for medical devices. Proven expertise in leading cross-functional teams to meet critical project milestones. Proficient in MS Office Suite, SolidWorks, Minitab, and ISO Compliance. Eager to leverage my technical proficiency and innovative mindset to improve healthcare outcomes broadly.

PROFESSIONAL EXPERIENCE

Prime Healthcare at St. Clare's Hospital – Denville, New Jersey
BioMedical Engineer I, BioMed Services
June 2025 – Present

- **Evaluate and optimize existing medical equipment performance**, developing and implementing test methods to ensure safety, efficacy, and compliance with hospital and regulatory standards.
- **Lead troubleshooting and facilitate repair of complex medical equipment**, including electro-mechanical issues, and complete timely maintenance, reducing equipment downtime and ensuring consistent operational standards.
- **Ensure thorough documentation of feedback from medical personnel regarding equipment performance**, creating comprehensive reports that enable cross-functional engineering teams to identify root causes and implement fixes in clinical settings.

Career Hiatus for Family Focus – Totowa, New Jersey
October 2019 – June 2025

Becton, Dickinson and Company – Franklin Lakes, New Jersey *[total tenure January 2013 – October 2019]*
Engineer II, Test Method Development, Medication Delivery Solutions, R&D
May 2017 – October 2019

- **Drove test method innovation**, designing and validating test methods for unique CSTD components, ensuring compliance with design verification/validation standards, and facilitating manufacturing scale-up.
- **Reduced testing requirements by 50%**, leveraging DOE to optimize critical dimensions on test fixtures, boosting efficiency and cutting costs without compromising quality.
- **Engineered custom test fixtures** using SolidWorks, integrating 3D metal printing and SLA capabilities to meet rigorous testing requirements.
- **Led global collaboration** to develop a next-generation closed system drug transfer (CSTD) device, enhancing hazardous drug protection for healthcare workers.

Shelf-Life Verification Process Owner / Engineer II
November 2014 – May 2017

- **Led a multi-site, cross-functional team** to reduce a significant backlog of shelf-life verification documents under a CAPA initiative, successfully achieving on-time completion.
- **Championed a Six Sigma Green Belt project** to pinpoint and eliminate root causes of delays in the Shelf-Life Verification Process, improving efficiency and boosting efficacy while streamlining risk assessment and escalation workflows to tackle product performance concerns head-on.
- **Owned technical review and approval** of product shelf-life verification reports, ensuring content integrity, and compliance with design verification protocols and product specification requirements.

- **Delivered actionable progress updates** and resource forecasts to senior leadership, keeping communication consistent, transparent, and aligned with company goals.

Engineer I / Shelf-Life Verification Engineer
January 2013 – November 2014

- Performed statistical analysis on real-time aging test results and authored product shelf-life verification reports for design history file (DHF) attachment and NSAI submission for CE Mark renewal.
- Developed technical rationale(s) with a cross-functional team to address MPS product performance concerns to ensure risk was mitigated when identified in shelf-life verification data.
- Authored / Updated Measurement System Analysis (MSA) Protocols to provide accurate guidelines to qualify and validate changes to Test Instructions; Authored MSA Reports after completion of gage R&R or equivalency study testing to approve the qualification and validation of changes to Test Instructions.
- Supported R&D efforts for medication delivery solutions, focusing on mechanical testing and materials property analysis to ensure product reliability.

Microwize Technology – Paramus, New Jersey
EHR Implementation Specialist, Allscripts MyWay
November 2011 – May 2012

- Led training and provided support to healthcare professionals when implementing and using Allscripts MyWay EHR software.
- Collaborated with medical teams to assess workflows and optimize software utilization for efficiency.
- Advised physicians on the CMS Incentive Program, ensuring compliance and maximizing (monetary) benefits.

Celgene Cellular Therapeutics – Warren, New Jersey
Intern, Cell Process Development
June 2009 – September 2009

- Evaluated microcarrier culture feasibility for large-scale mesenchymal stem cell production, contributing to process optimization for therapeutic applications.

EDUCATION

New Jersey Institute of Technology – Newark, NJ
Master of Science in Biomedical Engineering
Concentration: Biomechanics and Rehabilitation Engineering | GPA: 3.40 | May 2011

- NJIT Master's Fellowship Recipient

Stevens Institute of Technology – Hoboken, NJ
Bachelor of Engineering in Biomedical Engineering
Minor: Social Science | GPA: 3.20 | May 2009

- Dean's List | Women in Engineering Scholarship

KEY PROJECTS

Posterior Cruciate Ligament Tensioning Device – Stevens Institute of Technology
Senior Design Project Leader | August 2008 – May 2009

- Designed and prototyped a medical device to optimize PCL tensioning during reconstructive surgery; composed, presented, and filed a provisional patent.
- Awarded 2nd Place at Stevens Research and Entrepreneurship Day Elevator Pitch Olympics.

KATHERINE A. BARBIERI

TECHNICAL SKILLS

- *Software*: Bluehill Materials Testing Software, Microsoft Office Suite, Minitab, SolidWorks
- *Laboratory*: Mechanical Strength Testing, Pressure/Aspiration Testing (ISO 594-1/2, ISO 7886:1), Cell Culture (co-culture, scale-up), Motion Analysis

AWARDS & RECOGNITION

- 1st Place, 2009 Annual Meeting of the International Society for Pharmaceutical Engineering *Evaluation of Posterior Cruciate Ligament Tensioning Device.* San Diego, CA: November 2009.
- 2nd Place, Stevens Research and Entrepreneurship Day Elevator Pitch Olympics
- *Evaluation of Posterior Cruciate Ligament Tensioning Device.* Hoboken, NJ: May 8, 2009.
- NJIT Master's Fellowship (2011)
- Women in Engineering Scholarship, Stevens Institute of Technology (2009)

PROFESSIONAL DEVELOPMENT

- President, BD Speaks (Toastmasters International)
- Member, Healthcare Businesswoman's Association (HBA)
- Member, Society for Women Engineers (SWE)
- Member-at-Large, Delta Phi Epsilon Sorority

PUBLICATIONS

Evaluation of Posterior Cruciate Ligament Tensioning Device

- 6th Annual NJ Biomedical Engineering (BME) Showcase, NJIT, March 2009 (Poster)
- 35th Annual Bioengineering Conference, Harvard – MIT, April 2009 (Abstract & Poster)
- Senior Design Day, Stevens Institute of Technology, April 2009 (Poster)

CORE COMPETENCIES

Design Control
Technical Writing
Microsoft Office Suite
Quality Management Systems
Cross-Functional Team Leadership
Minitab Statistical Analysis Software
Medical Device Design and Prototyping
SolidWorks and 3D Printing (SLA, Metal)
Test Method Development and Validation
Design of Experiments (DOE) and Six Sigma
21 CFR Part 820: Quality System Regulation
Biomechanics and Rehabilitation Engineering
Regulatory Body Report Submission (FDA / NSAI)
Measurement System Analysis (MSA) and Gage R&R
ISO 7886-1: Sterile Hypodermic Syringes for Single Use
ISO 134858: Medical Devices – Quality Management Systems
ISO 80369-7: Small-bore Connectors for Liquids and Gases in Healthcare Applications (ISO 594-1/2)